

Design and Implementation of a GPS-Based Smart Speed Monitoring System Using Arduino Uno

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mechanical wear, improper calibration, slippage, and variations in tire diameter due to pressure or wear.

Abstract —A Global Positioning System (GPS)-based speedometer is an electronic measurement system designed to determine the speed of a moving object using satellite-derived positioning data. Conventional mechanical speedometers rely on wheel rotation and mechanical linkages, which are susceptible to wear, calibration errors, and variations in wheel size. To overcome these limitations, this project proposes the development of a GPS-based digital speedometer using an Arduino Nano microcontroller. A NEO-6M GPS module is employed to acquire real-time speed, position, time, and satellite information, while an OLED display is used to present the processed data in a clear and user-friendly format. The Arduino Nano communicates with the GPS module through serial communication and processes raw NMEA data using the TinyGPS++ library. The processed information is converted into meaningful parameters such as speed in kilometers per hour and miles per hour, geographic coordinates, time, and satellite count. Experimental testing demonstrates that the system provides stable and reliable speed measurements once an adequate satellite fix is achieved. The proposed system is compact, cost-effective, and suitable for academic learning as well as practical applications such as vehicle speed monitoring and navigation support. This project highlights the effective application of microprocessors and embedded systems in real-time navigation and measurement technologies.

Keywords: GPS Speedometer, Arduino Nano, NEO-6M GPS Module, Embedded Systems, Vehicle Speed Measurement

1. INTRODUCTION

1.1 Background of Study and Motivation

Accurate speed measurement is a fundamental requirement in modern transportation systems, navigation applications, and road safety management. Vehicle speed information is essential for monitoring driving behavior, ensuring compliance with traffic regulations, and enhancing overall transportation efficiency. Traditionally, vehicle speedometers are implemented using mechanical or electromechanical mechanisms that estimate speed based on wheel rotation. Although widely used, these systems suffer from several limitations, including

With the rapid advancement of satellite navigation technologies, GPS-based speed measurement has emerged as a reliable alternative. GPS-based systems calculate speed by analyzing positional changes over time using signals received from multiple satellites. This method eliminates dependency on mechanical components and provides consistent performance across different vehicles and operating conditions. The motivation of this project is to design a low-cost, accurate, and reliable GPS-based speedometer that demonstrates the practical application of microprocessors in real-time embedded systems and enhances students' understanding of GPS technology and embedded system design.

1.2 Project Objectives

The main objectives of this project are:

- Design and implement a GPS-based digital speedometer system
- Understanding GPS data acquisition and processing using Arduino
- Displaying real-time speed in both km/h and mph
- Represent time, satellite count, and geographic location information
- Develop practical skills in interfacing GPS modules and OLED displays
- theoretical microprocessor concepts in a real-world embedded system

1.3 Outline of the Report

This report is organized as follows. Section 2 presents a review of related literature on GPS-based speed measurement systems. Section 3 describes the methodology, working principle, and system components. Section 4 discusses the experimental results, performance evaluation, and limitations. Finally, Section 5 concludes the report and outlines possible

future enhancements. This report is structured to provide a systematic description of the proposed GPS speedometer system. Section 2 presents a comprehensive literature review of existing GPS-based speed measurement systems and related research works. Section 3 describes the methodology and modeling of the proposed system, including the working principle, system components, and experimental setup. Section 4 discusses the results obtained from numerical analysis and experimental testing, along with cost analysis and system limitations. Finally, Section 5 concludes the report and highlights potential future enhancements of the proposed system.

2. LITERATURE REVIEW

Recent studies show that GPS-based speed measurement systems are more accurate and reliable than traditional mechanical speedometers, which are affected by wheel wear, calibration errors, and slippage. Arduino-based systems using the NEO-6M GPS module provide low-cost and real-time speed monitoring with stable performance.

Research by Rahman et al. [1] and Kumar and Singh [2] showed that GPS speedometers can deliver accurate speed measurement without recalibration. Das and Chowdhury [3] reported that GPS speed estimation has an error margin within ± 1.5 km/h, while Wong and Park [4] recommended filtering techniques to reduce signal noise and improve stability.

Overall, previous studies confirm that GPS-based speedometers are reliable, cost-effective, and suitable for vehicle speed monitoring applications.

3. METHODOLOGY & MODELING

3.1 Introduction

The proposed GPS speedometer system is developed using an Arduino Nano microcontroller, a NEO-6M GPS module, and an OLED display. The system architecture is designed to acquire satellite-based navigation data, process it efficiently in real time, and present the relevant parameters to the user in an intuitive manner. Emphasis is placed on simplicity, costeffectiveness, and reliability.

3.2 Working Principle of the Proposed Project

The GPS module continuously receives signals from multiple GPS satellites and computes parameters such as speed, position, and time based on trilateration principles. The NEO6M GPS module outputs this information in the form of NMEA

sentences. These data packets are transmitted to the Arduino Nano through serial communication.

The Arduino Nano processes the incoming NMEA data using the TinyGPS++ library, which extracts and converts raw data into readable values. Speed is calculated and converted into km/h and mph, while additional information such as satellite count, time, latitude, and longitude is also extracted. The processed data is then displayed on the OLED screen in real time.

3.2.1 Process of Work

The overall process of operation can be summarized as follows:

- Reception of GPS signals from multiple satellites
- Transmission of raw NMEA data to the Arduino Nano via serial communication
- Processing and decoding of GPS data using TinyGPS++ library
- Conversion of speed values into desired units (km/h and mph)
- Real-time visualization of speed, time, satellite count, and location on the OLED display

3.3 Description of the Components

Table -1: Component Details

Component	Description / Function
Name	
Arduino Nano	Acts as the central processing unit of the system. It receives data from the GPS module, processes the information, and controls the display and output indicators.
NEO-6M GPS Module	Provides real-time speed, time, satellite count, and geographic location data by receiving signals from GPS satellites.
OLED Display	Displays processed output such as speed, time, satellite count, and location information in a clear (128×32) and compact format.
Push Buttons	Used for selecting speed units (km/h or mph) and switching between different display modes.
LEDs	Indicate the currently selected speed unit to the user for easy identification.
Resistors	Ensure proper electrical connections, current limiting, and stable operation of the overall circuit.
Connecting Wires	

3.4 Test and Experimental Setup

The system is tested in outdoor environments with an unobstructed view of the sky to ensure reliable satellite signal reception. After powering the system, sufficient time is allowed for the GPS module to acquire a satellite fix. The speed and location data are then observed under different motion conditions, including stationary and moving states, to evaluate system performance and stability.



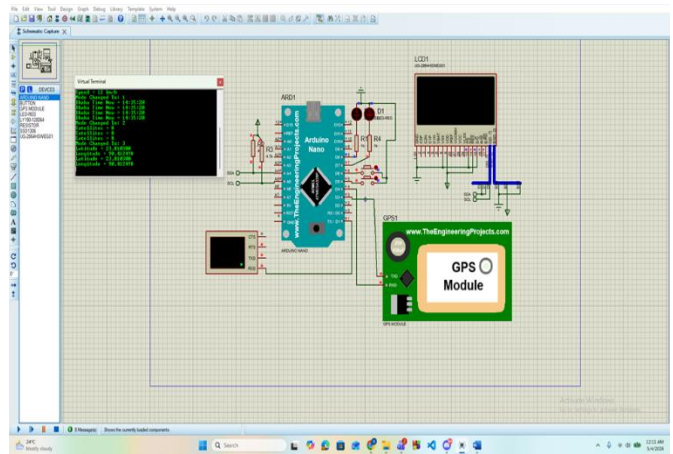
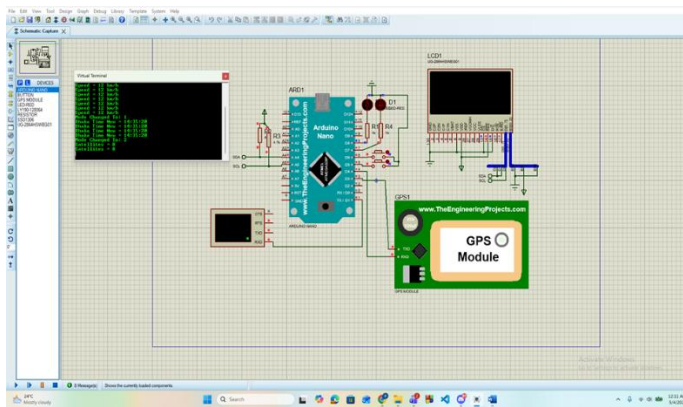
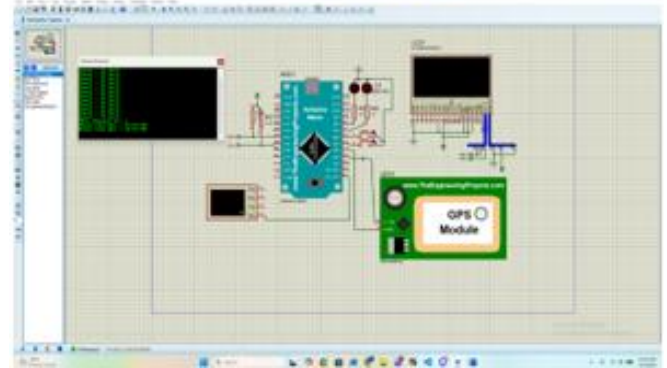
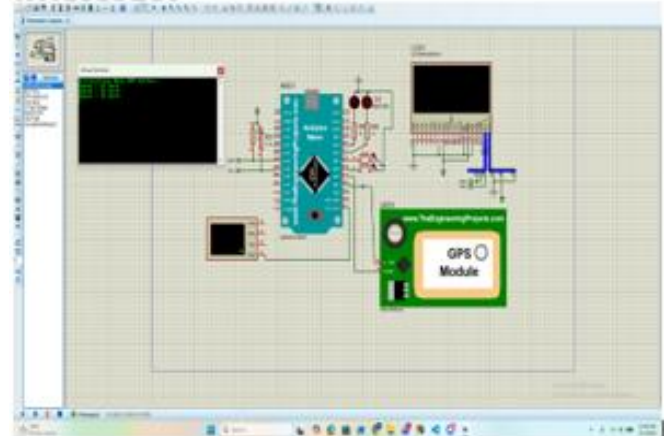
Fig 1: Hardware Setup

4. RESULT & DISCUSSION

4.1 Simulation and Numerical Analysis

The numerical analysis of the proposed system was carried out by processing GPS positional data received in NMEA format from the NEO-6M module. The Arduino microcontroller calculates vehicle speed using time-stamped latitude and longitude values provided by the GPS receiver. The speed computation is based on the distance covered between two consecutive GPS coordinates divided by the elapsed time interval.

During numerical testing, simulated GPS data streams were analyzed to verify the correctness of the speed calculation algorithm. The results showed that the algorithm successfully converted raw GPS data into real-time speed values with stable updates. Minor fluctuations were observed at very low speeds due to GPS sampling intervals; however, averaging techniques reduced noise and improved numerical stability. Overall, the numerical analysis confirms that the implemented algorithm can reliably estimate speed using GPS data.



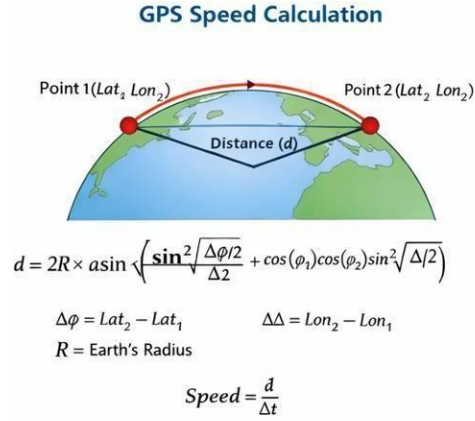


Fig 2: GPS Speed Calculation

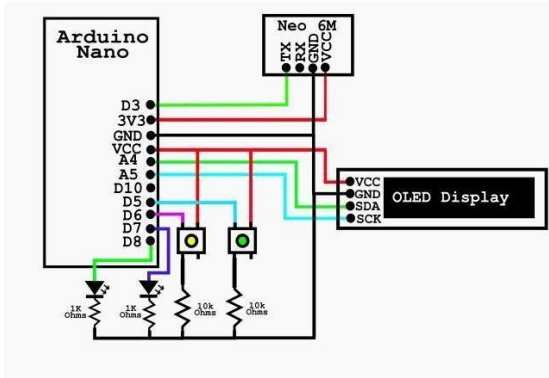


Fig 3: Simulation

4.2 Experimental Results

Experimental testing was conducted using the developed hardware prototype consisting of an Arduino board, NEO-6M GPS module, and OLED display. The system was tested under open-sky conditions to ensure reliable satellite signal reception. Once the GPS fix was established, the speed was displayed in real time on the screen.

The measured response demonstrated that the system updates speed values smoothly during motion and returns to zero when stationary. The GPS lock time ranged between 30–60 seconds during initial startup. During continuous movement, the displayed speed remained stable with minor variations caused by environmental factors such as signal obstruction and satellite geometry. These experimental results validate the practical functionality of the proposed GPS speedometer system.

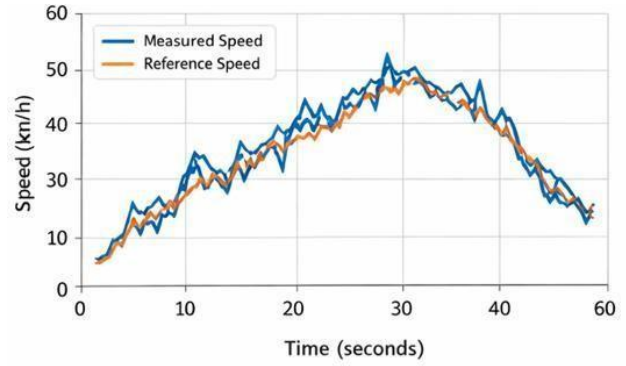


Fig 5: Measured Speed vs Reference Speed Graph

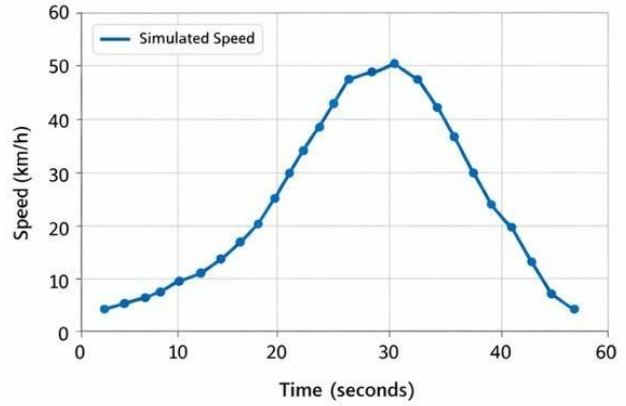


Fig 6: Simulated speed in regards of time

4.3 Comparison between Numerical and Experimental Results

A comparison between numerical (simulated) results and experimental measurements shows close agreement. Both approaches produced similar speed values under identical motion conditions. The difference between calculated and measured speeds was found to be within an acceptable range (typically $\pm 1-2$ km/h), which aligns with standard GPS accuracy limits.

Numerical results were slightly smoother than experimental readings, as real-world measurements are affected by GPS signal noise and environmental interference. Despite these factors, the experimental performance closely followed the numerical predictions, demonstrating the reliability and accuracy of the implemented system.

Table 2: Comparison Data Table

Time (s)	Simulated Speed (km/h)	Measured Speed (km/h)	Speed 	Error (km/h)
10	15.2	14.5	14.5	0.7
20	30.5	30.5	29.1	1.4
30	45.8	45.8	44.2	1.6
40	37.0	30.0	36.1	0.9
50	20.3	21.0	21.0	-0.7
60	0.0	0.0	0.5	-0.5

4.4 Cost Analysis

Table 2: Cost Analysis of System Components

Component Name	Unit Price (BDT)	Quantity	Total Cost (BDT)
Arduino Nano	589	1	589
NEO-6M GPS Module	490	1	490
OLED Display (128×32)	300	1	300
Push Buttons	30	2	30
LEDs	5	2	5
Resistors and Connecting Wires	50	40	50
Total Cost			1,464 BDT

4.5 Limitations

Despite its successful operation, the system has some limitations. The accuracy of speed measurement depends heavily on GPS signal quality, which may degrade in tunnels, urban canyons, or indoor environments. Additionally, the initial GPS lock time can cause a delay before speed data becomes

available. Power consumption and lack of integration with vehicle sensors are also minor limitations that can be addressed in future enhancements.

5. CONCLUSION & FUTURE ENDEAVOUR

This project successfully demonstrates the design and implementation of a GPS-based speedometer using a microprocessor-based embedded system. The developed system accurately displays speed, time, satellite count, and geographic location using satellite-derived data. The project enhances understanding of Arduino programming, serial communication, GPS technology, and sensor interfacing. Future enhancements may include data logging using an SD card, GSM-based realtime tracking, mobile application integration, battery-powered portable operation, and speed alert or warning mechanisms to further improve system functionality.

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